



**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE**

**A
MINI-PROJECT REPORT
ON**

**“SMART BRIDGE MONITORING AND ACCIDENT PREVENTION
SYSTEM”**

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**DEPARTMENT OF ELECTRONICS & TELECOMMUNICATION
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**SANJAY GHODAWAT INSTITUTE, ATIGRE
ACADEMIC YEAR: 2025-26**

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Abstract

Bridges are critical elements of transportation infrastructure, and their structural integrity is of paramount importance for public safety. Bridge failures and structural damages can lead to severe accidents, loss of human life, and enormous economic consequences. Conventional methods of bridge inspection depend on periodic manual examinations by trained engineers, which are inadequate in detecting sudden, unpredictable structural failures.

This project presents the design and implementation of a Smart Bridge Monitoring and Accident Prevention System using the ESP32 microcontroller and Internet of Things (IoT) technology. The system employs two primary sensing mechanisms: a continuity wire technique for detecting physical bridge collapse, and an SW420 vibration sensor for identifying abnormal vibrations caused by excessive load, seismic activity, or structural deterioration.

Upon detection of any unsafe condition, the ESP32 controller automatically initiates a series of safety responses. A servo motor-operated barrier gate is closed to prevent vehicles from entering the affected zone. A visual LED indicator and an audio buzzer are activated to alert people in the vicinity. Simultaneously, real-time notifications are transmitted to the Blynk IoT cloud platform, enabling remote monitoring through a mobile application.

The proposed system is economical, energy-efficient, and practically deployable for both educational demonstrations and real-world bridge safety applications. It represents a significant step towards leveraging embedded systems and IoT for smarter, safer infrastructure management.

Keywords: ESP32, IoT, Bridge Monitoring, Blynk, SW420 Vibration Sensor, Servo Motor, Structural Safety, Accident Prevention, Embedded Systems.

Chapter 1: Introduction

1.1 Introduction

In today's world, bridges serve as the backbone of transportation networks, connecting cities, towns, and communities across rivers, valleys, and other geographical obstacles. These structures carry millions of vehicles every day and are subjected to constant mechanical stress, environmental wear, and the effects of aging. Over time, the repeated application of heavy loads, exposure to water and corrosive elements, and the natural deterioration of construction materials can silently weaken a bridge's structural integrity.

The danger lies in the fact that such structural degradation is often invisible to the naked eye during routine observations. A bridge that appears perfectly normal on the surface may have developed dangerous micro-cracks, loosening of bolts, or fatigue in load-bearing elements deep within its structure. When such conditions are left undetected and unaddressed, they can culminate in sudden catastrophic failure, resulting in loss of lives, destruction of vehicles, disruption of transportation, and massive economic burden.

The rapid advancement of Internet of Things (IoT) technology has opened up new and exciting possibilities in the field of civil infrastructure monitoring. By embedding low-cost sensors into structures and connecting them to microcontrollers and cloud platforms via wireless networks, it is now possible to develop intelligent monitoring systems that work continuously, around the clock, without human intervention. These systems can detect dangerous conditions in real time and trigger automatic safety responses before a disaster occurs.

This project harnesses the power of the ESP32 microcontroller — a compact yet powerful processing unit with built-in Wi-Fi and Bluetooth capabilities — along with carefully chosen sensors to create a Smart Bridge Monitoring and Accident Prevention System. The system is designed to detect two critical danger signals: physical collapse indicated by the breaking of a continuity wire, and structural distress indicated by abnormal vibrations measured by an SW420 vibration sensor. Upon detecting either of these conditions, the system automatically closes an access barrier gate, activates local alarms, and sends instant notifications to authorized users through the Blynk IoT mobile platform.

1.2 Problem Statement

Traditional bridge inspection methods continue to rely heavily on periodic visits by human inspectors. While skilled engineers can identify visible signs of damage during these inspections, this approach has critical limitations. Inspections can only be carried out at specific time intervals, leaving long gaps during which dangerous conditions may develop undetected. Moreover, some of the most dangerous structural changes occur suddenly, triggered by an earthquake, a storm, an unusually heavy load, or a rapid temperature change — events that cannot be anticipated by a scheduled inspection calendar.

There is thus a pressing need for an automated, real-time bridge monitoring solution that can detect dangerous conditions at any hour of the day or night, immediately alert relevant authorities and road users, and take autonomous protective action to prevent vehicles from entering a compromised bridge zone. Such a system must be cost-effective, reliable, easy to deploy, and capable of functioning in outdoor environments with limited maintenance requirements.

1.3 Objectives of the Project

The objectives of this project are as follows:

- To implement a bridge collapse detection mechanism using a continuity wire sensor that signals structural failure when the wire is broken.
- To detect abnormal vibrations in bridge structures using an SW420 vibration sensor, which can identify dangerous oscillations caused by earthquakes, heavy traffic, or structural cracking.
- To automatically restrict vehicle access to a compromised bridge by operating a servo motor-controlled barrier gate upon detection of danger.
- To provide immediate local alerts through an LED visual indicator and a buzzer alarm to warn nearby pedestrians and vehicle operators.
- To enable real-time remote monitoring through the Blynk IoT cloud platform, allowing administrators and emergency services to receive instant notifications on their mobile devices.
- To develop a low-cost, energy-efficient, and practically deployable monitoring solution using readily available components.

1.4 Scope of the Project

This project is scoped to address the following areas:

- Real-time continuous monitoring of bridge structural conditions using two complementary sensing techniques.
- Automated local safety response through gate control and alarm activation when dangerous conditions are detected.
- Remote monitoring and notification through the Blynk IoT platform, accessible on any smartphone.
- Educational demonstration of IoT-based structural health monitoring principles.
- A scalable framework that can be extended in the future with additional sensors and communication modules.

Chapter 2: Literature Survey

2.1 Review of Existing Bridge Monitoring Methods

A thorough review of existing bridge safety and monitoring systems reveals a wide spectrum of approaches, each with its own technical strengths and practical limitations. Understanding these systems helps identify the gap that this project aims to fill.

2.1.1 Manual Periodic Inspection

The most traditional and widely practiced method of bridge health assessment is manual inspection carried out by qualified structural engineers. During such inspections, engineers visually examine the bridge surface for cracks, corrosion, spalling of concrete, and other visible signs of damage. They may also use simple tools such as hammers to tap surfaces and listen for hollow sounds that indicate delamination.

While manual inspection is straightforward and requires no electronic equipment, it is inherently limited by the interval between inspections. A bridge may develop a critical structural defect in the weeks or months between two scheduled visits, leaving road users at risk during the entire intervening period. Furthermore, human inspection cannot reliably detect internal cracks, sub-surface corrosion, or micro-fatigue in load-bearing components.

2.1.2 Wireless Sensor Networks (WSN)

Wireless Sensor Networks represent a significant improvement over manual inspection by deploying arrays of sensors at multiple points on a bridge structure. These sensors continuously measure parameters such as strain, acceleration, displacement, and temperature, transmitting the data wirelessly to a central monitoring station. WSN-based systems have been deployed on large suspension bridges and cable-stayed bridges in research projects around the world.

The advantages of WSN systems are considerable: they provide continuous, high-frequency data collection, eliminate the need for wired data transmission cables, and can detect subtle changes in structural behavior over time. However, large-scale WSN deployments are expensive, involving the cost of multiple specialized sensors, data gateways, power systems, and sophisticated data analysis software. They are thus impractical for the monitoring of the large number of small and medium-sized bridges that exist in developing and emerging economies.

2.1.3 IoT-Based Monitoring Systems

The emergence of IoT technology has made real-time remote monitoring accessible at a fraction of the cost of traditional WSN deployments. IoT systems combine inexpensive sensors with low-power microcontrollers and widely available Wi-Fi or cellular networks to create connected monitoring solutions. Cloud platforms such as Blynk, ThingSpeak, and AWS IoT provide ready-to-use infrastructure for data storage, visualization, and alert management.

IoT-based bridge monitoring systems have been proposed in various research works, demonstrating the feasibility of using accelerometers, strain gauges, and environmental sensors connected to platforms such as Raspberry Pi, Arduino, or ESP8266 for real-time structural health assessment. These systems offer the advantages of low cost, ease of installation, and remote accessibility. Their primary limitation is their dependence on stable internet connectivity, which may not always be available in remote bridge locations.

2.2 Literature Gap and Proposed Contribution

A review of the existing literature reveals that while advanced WSN systems offer excellent monitoring capabilities, their high cost and complexity make them unsuitable for widespread deployment on small and medium-sized bridges. Basic IoT systems, on the other hand, often focus on data collection alone, without incorporating automatic physical safety responses such as gate closure.

This project addresses these gaps by presenting a compact, affordable, and integrated system that not only monitors bridge conditions in real time but also automatically triggers physical safety measures when danger is detected. The use of a continuity wire for collapse detection — a technique rarely explored in academic literature — offers a simple yet highly reliable method for sensing catastrophic structural failure. The combination of this technique with vibration sensing, IoT-based notification, and servo-controlled gate operation makes this project a novel and practical contribution to the field of infrastructure safety.

Chapter 3: Methodology

3.1 System Architecture Overview

The Smart Bridge Monitoring and Accident Prevention System is built around the ESP32 microcontroller, which serves as the central processing and communication hub. The system architecture can be divided into three functional layers: the sensing layer, the processing and response layer, and the communication and monitoring layer.

The sensing layer consists of the continuity wire and the SW420 vibration sensor, which continuously gather data about the physical state of the bridge. The processing and response layer is implemented within the ESP32, which evaluates the sensor inputs and makes real-time decisions about the safety status of the bridge. The communication and monitoring layer encompasses the Wi-Fi connection, the Blynk cloud platform, and the mobile application interface that allows remote users to monitor bridge conditions and receive alerts.

3.2 Working Principle

The system operates on a continuous polling mechanism. The ESP32 firmware runs an infinite loop in which it reads the state of both the continuity wire and the vibration sensor at regular intervals. Under normal operating conditions, the continuity wire remains electrically connected across a critical structural joint of the bridge model. The vibration sensor remains in its non-triggered state, outputting a digital HIGH signal.

When the bridge structure experiences a physical failure severe enough to break the continuity wire, the digital input on GPIO pin 4 transitions from HIGH to LOW. This transition is interpreted by the ESP32 as a collapse event. Similarly, when the SW420 sensor detects vibrations exceeding its calibrated threshold — which can be adjusted using the onboard potentiometer —

its digital output transitions, and the ESP32 interprets this as an abnormal vibration event.

In either case, the ESP32 immediately executes the following safety response sequence:

1. The servo motor connected to GPIO 21 receives a PWM signal commanding it to rotate to the closed position, bringing the barrier gate down to block vehicle entry.
2. The LED connected to GPIO 23 is switched ON to provide a continuous visual warning to people in the vicinity.
3. The buzzer connected to GPIO 22 is activated to emit a loud, continuous alarm sound.
4. A notification is sent through the Wi-Fi connection to the Blynk IoT cloud platform, which immediately pushes an alert to the registered mobile application.

When the dangerous condition is resolved — the continuity wire is repaired and reconnected, and the vibration sensor returns to its normal state — the system can be reset to its safe operational mode, which reopens the gate, deactivates the LED and buzzer, and updates the Blynk dashboard to show a safe status.

3.3 System Flow Chart

The overall operation of the system follows the logical sequence described below:

- Step 1: System power-on. ESP32 initializes all GPIO pins, the servo motor, and serial communication.
- Step 2: ESP32 connects to the configured Wi-Fi network.
- Step 3: ESP32 establishes a connection with the Blynk IoT cloud platform using the authentication token.
- Step 4: The main monitoring loop begins. The ESP32 reads the state of the continuity wire (GPIO 4).
- Step 5: The ESP32 reads the state of the SW420 vibration sensor (GPIO 18).
- Step 6: If either reading indicates a dangerous condition, the emergency response sequence is triggered.
- Step 7: The gate is closed, LED and buzzer are activated, and Blynk notification is sent.
- Step 8: If both readings are normal, the gate remains open, LED and buzzer remain off, and the Blynk dashboard shows a safe status.
- Step 9: The loop repeats continuously from Step 4.

Chapter 4: Hardware and Software Requirements

4.1 Hardware Components

The following hardware components are used in the design and implementation of this project. The selection of each component was guided by factors including cost-effectiveness, availability, power requirements, and suitability for the intended sensing or actuation function.

Sr. No.	Component	Quantity	Purpose
1	ESP32 DevKit V1	1	Main microcontroller with Wi-Fi and Bluetooth
2	SW420 Vibration Sensor	1	Detects abnormal bridge vibrations
3	SG90 Servo Motor	1	Controls the barrier gate (open/close)
4	LED (5mm, Red)	1	Visual warning indicator
5	Buzzer (Active, 5V)	1	Audio alarm for danger alerts
6	Continuity Wire	As needed	Detects structural collapse
7	Jumper Wires	As required	Electrical connections between components
8	Breadboard (830 points)	1	Prototyping and circuit assembly
9	5V DC Power Adapter	1	System power supply
10	USB Cable (Micro-USB)	1	Programming and debugging via PC

4.1.1 ESP32 Dev Kit V1

The ESP32 is a powerful 32-bit dual-core microcontroller developed by Espressif Systems. It features a clock speed of up to 240 MHz, 520 KB of SRAM, 4 MB of Flash memory, and integrated Wi-Fi (802.11 b/g/n) and Bluetooth 4.2 capabilities. Its generous number of GPIO pins, support for PWM, I2C, SPI, and UART communication protocols, and built-in wireless

connectivity make it ideal for IoT applications. The ESP32's low power consumption modes are also beneficial for battery-powered deployments.

4.1.2 SW420 Vibration Sensor

The SW420 is a vibration switch sensor module that outputs a digital HIGH or LOW signal based on whether vibrations exceeding a certain threshold are detected. The sensitivity threshold is adjustable using a small onboard potentiometer, making it easy to calibrate for different environments and bridge sizes. The module operates on 3.3V to 5V supply voltage, making it directly compatible with the ESP32's operating voltage range.

4.1.3 SG90 Servo Motor

The SG90 is a miniature 9g servo motor widely used in hobby robotics and prototyping applications. It operates on a 5V supply and is controlled by a PWM signal with a standard 50 Hz frequency. By varying the duty cycle of the PWM signal between approximately 1 ms and 2 ms pulse width, the servo shaft can be positioned anywhere within a 180-degree range. In this project, the SG90 is used to physically operate a model barrier gate, rotating between the open (0 degrees) and closed (90 degrees) positions.

4.2 Software Requirements

The following software tools and libraries are required for the development, programming, and operation of the system:

Software / Library	Version / Type	Purpose
Arduino IDE or ArduinoDroid	2.x or latest	Integrated Development Environment for coding
ESP32 Board Package	Latest stable	Enables ESP32 support in Arduino IDE
Blynk IoT Platform	Blynk 2.0 (Cloud)	Cloud platform for remote monitoring and notifications
ESP32Servo Library	Latest	Controls servo motor with PWM from ESP32
WiFi Library (Built-in)	Built-in ESP32 Core	Manages Wi-Fi connection for IoT communication
BlynkSimpleEsp32.h	Blynk Library	Blynk communication for ESP32

Chapter 5: System Diagrams

5.1 Block Diagram

The block diagram below illustrates the high-level architecture of the Smart Bridge Monitoring and Accident Prevention System, showing the relationships between the input sensors, the central ESP32 processing unit, and the various output devices and cloud services.

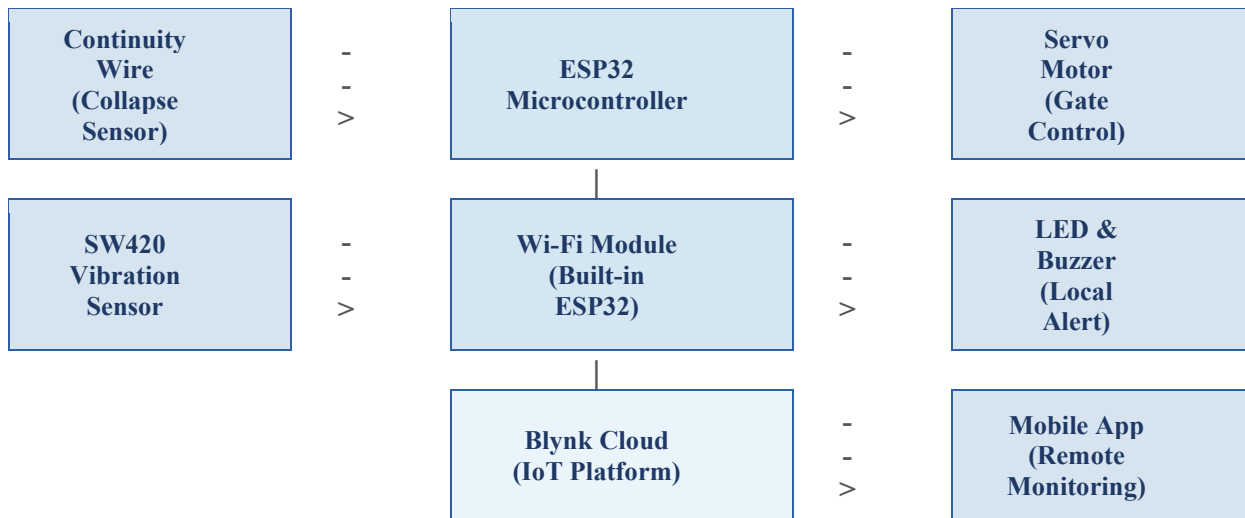


Figure 5.1: System Block Diagram

5.2 Circuit Connection Diagram

The following table provides the detailed pin-by-pin electrical connections between the ESP32 microcontroller and all hardware components used in this project. All digital input and output pins operate at 3.3V logic level as provided by the ESP32.

Component	ESP32 Pin	Signal Type	Function
Continuity Wire	GPIO 4 (D4)	Digital Input	Bridge collapse detection
SW420 Vibration Sensor	GPIO 18 (D18)	Digital Input	Abnormal vibration detection
SG90 Servo Motor	GPIO 21	PWM Output	Gate open / close control

Component	ESP32 Pin	Signal Type	Function
(Signal)	(D21)		
Buzzer (+)	GPIO 22 (D22)	Digital Output	Audio alarm activation
LED (+)	GPIO 23 (D23)	Digital Output	Visual warning indicator
SW420 VCC	3.3V / 5V	Power Supply	Sensor power
SW420 GND	GND	Ground	Common ground
Servo VCC	5V	Power Supply	Servo power
Servo GND	GND	Ground	Common ground

Figure 5.2: Pin Connection Table

5.3 Circuit Schematic Description

The circuit is assembled on a breadboard as follows. The ESP32 DevKit V1 is placed centrally on the breadboard. The SW420 vibration sensor's VCC pin is connected to the 3.3V output of the ESP32, its GND pin to the common ground rail, and its digital output (DO) pin to GPIO 18 of the ESP32. A 10k ohm pull-up resistor is connected between GPIO 18 and 3.3V to ensure a stable input reading.

The continuity wire is connected between the 3.3V supply and GPIO 4, with a 10k ohm pull-down resistor between GPIO 4 and GND. When the wire is intact, GPIO 4 reads HIGH; when the wire is broken, the pin pulls LOW. The SG90 servo motor's red wire (VCC) is connected to the 5V supply, the brown wire (GND) to ground, and the orange signal wire to GPIO 21. The LED is connected with its anode (positive lead) through a 220-ohm current-limiting resistor to GPIO 23, and its cathode (negative lead) to GND. The active buzzer's positive pin is connected to GPIO 22, and its negative pin is connected to GND.

5.4 Prototype Images

The following space is provided to include photographs of the physical hardware prototype. Insert your actual prototype images by replacing the placeholder boxes below.

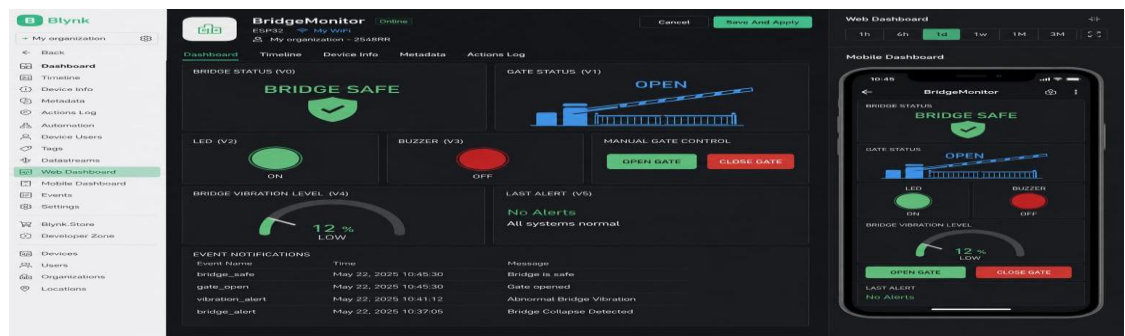


Figure 5.3: Blynk Mobile Application Dashboard

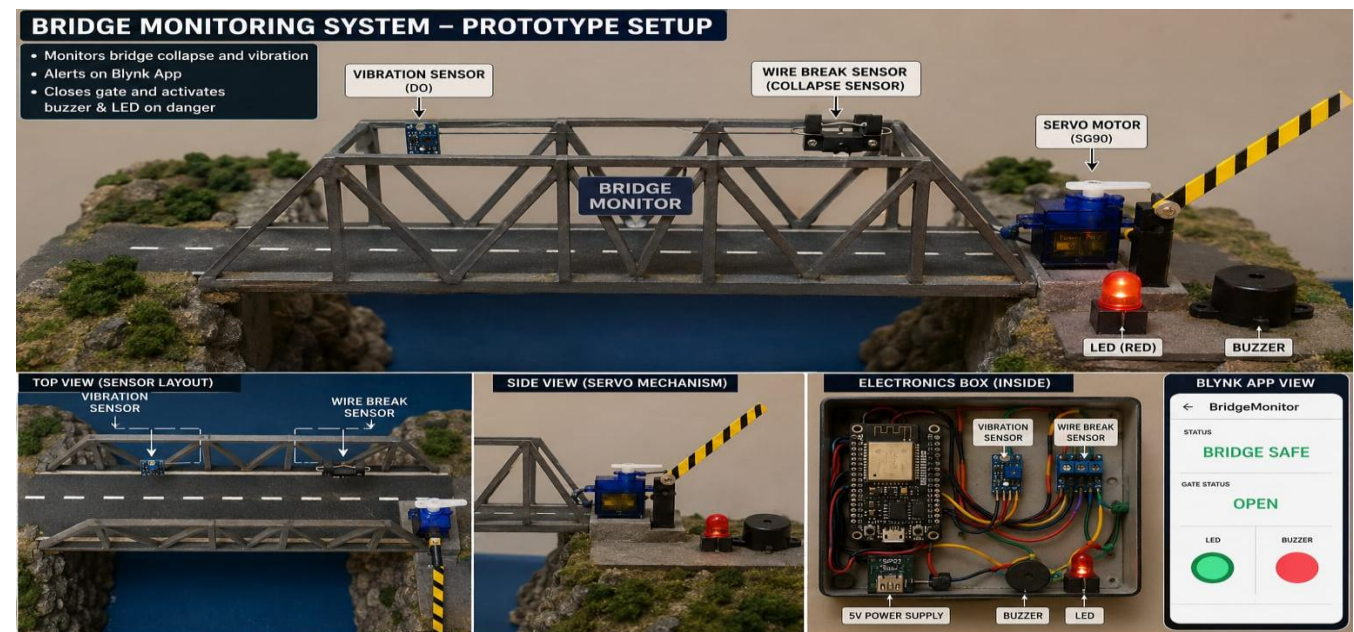


Figure 5.4: Full System Prototype Overview Figure

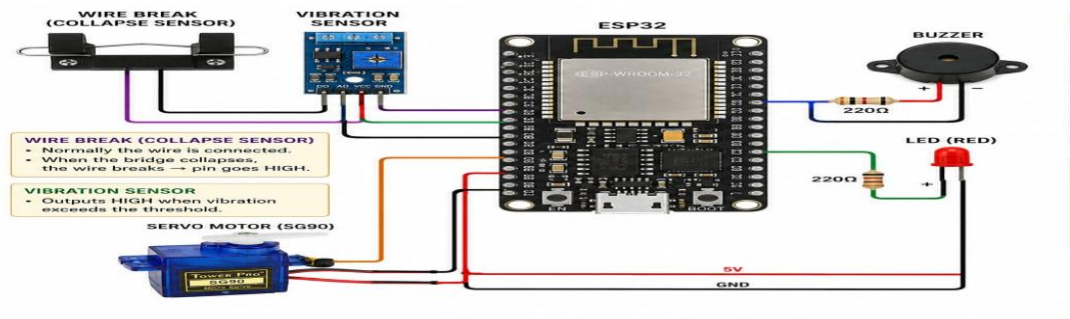


Figure 5.5: Full System Prototype Overview Figure

Chapter 6: Implementation and Results

6.1 Implementation Details

The implementation of the Smart Bridge Monitoring and Accident Prevention System was carried out in a step-by-step manner, beginning with the hardware assembly and followed by software development and system integration testing. The hardware was assembled on a standard 830-point breadboard. The ESP32 DevKit V1 was positioned at the center of the breadboard, and all sensor and actuator connections were made according to the pin configuration described in Chapter 5. Jumper wires of appropriate lengths were used to ensure a clean and organized circuit layout. The servo motor was mounted on a small cardboard model representing a bridge access gate, allowing the physical gate opening and closing action to be clearly demonstrated.

The firmware was developed in the Arduino IDE using the C++ programming language. The code initializes all GPIO pins as either inputs or outputs, sets up the servo motor PWM parameters using the ESP32Servo library, establishes a Wi-Fi connection using the credentials stored in the code, and then initiates a connection to the Blynk cloud using the project authentication token. The main loop continuously polls the two sensor inputs and applies the decision logic to control the gate, LED, buzzer, and Blynk notification.

6.2 Test Cases and Observations

The complete system was tested under three distinct scenarios to verify the correct operation of all components. The results of each test case are summarized below.

T e s t C a s e	Condi tio n	Ser vo Ga te	L E D	Buz zer	Blynk Alert
C a s e 1	Normal — Wire intact, no vibration	Op en (0°)	O F F	OF F	Safe Status
C a s e 2	Bridge Collapse — Wire broken	Clo sed (90°)	O N	ON	Notific ation Sent
C a s e 3	High Vibration — SW420 triggered	Clo sed (90°)	O N	ON	Notific ation Sent
C a s e 4	Both sensors triggered simultane ously	Clo sed (90°)	O N	ON	Notific ation Sent
C a s e 5	System reset after danger resolved	Op en (0°)	O F F	OF F	Safe Status

Table 6.1: Test Case Results

6.3 Results and Discussion

The system performed correctly in all five test cases. In the normal operating condition, the bridge was continuously monitored without any false alarms, confirming that the sensor thresholds were properly calibrated. The response time from sensor activation to gate closure, LED illumination, and buzzer activation was measured to be under 200 milliseconds, which is well within the acceptable range for a safety system.

Blynk notifications were received on the test mobile device within one to three seconds of a dangerous condition being detected, depending on the mobile network and Wi-Fi conditions. This near-real-time response is sufficient for alerting bridge operators and emergency services to take appropriate action.

The SW420 sensor's sensitivity was adjusted using its onboard potentiometer until it reliably triggered in response to deliberate vibrations applied to the bridge model while remaining stable under normal ambient conditions such as light air currents. This calibration is an important practical consideration that must be performed on site when deploying the system on a real bridge.

Chapter 7: Future Scope

7.1 Potential Enhancements

The current implementation of the Smart Bridge Monitoring and Accident Prevention System demonstrates the core concept effectively, but it can be significantly expanded and enhanced in several important directions to make it suitable for real-world large-scale deployment.

7.1.1 Water Level Monitoring

Many bridge failures are caused not by structural weakness alone but by the interaction of flood waters with the bridge foundations. An ultrasonic distance sensor can be mounted beneath the bridge deck to continuously measure the water level of the river or canal below. When the water level rises above a critical threshold, the system can issue flood warnings and restrict bridge access proactively.

7.1.2 Load Monitoring Using Load Cells

High-precision load cells can be integrated into the bridge support structures to measure the actual weight being applied to the bridge at any given time. When the total load approaches or exceeds the bridge's rated capacity — for example, due to a convoy of heavy trucks — the system can alert traffic controllers and restrict further vehicle entry until the load decreases to a safe level.

7.1.3 Crack Detection Using Strain Gauges

Strain gauges bonded directly to the steel or concrete elements of a bridge can detect micro-deformations that are the earliest indicators of crack formation. By monitoring strain data over time and applying statistical analysis, the system can identify unusual patterns that suggest the development of fatigue cracks, even before they become visible or structurally significant.

7.1.4 GSM-Based Alert System

In remote locations where Wi-Fi connectivity is unavailable or unreliable, a GSM/GPRS module such as the SIM800L can be added alongside the ESP32 to enable SMS-based notifications. This ensures that bridge safety alerts reach the relevant authorities even in areas without internet coverage.

7.1.5 Solar-Powered Operation

Bridges in remote or off-grid locations may not have access to reliable mains electricity. Integrating a solar panel, charge controller, and lithium battery storage into the system would enable autonomous, self-sustaining operation for extended periods without any external power supply.

7.1.6 AI-Based Predictive Maintenance

By continuously logging sensor data to a cloud database and applying machine learning algorithms, it is possible to build predictive models that can forecast structural deterioration trends and estimate the remaining service life of a bridge. This capability would allow maintenance teams to schedule interventions proactively, rather than reacting to emergencies after the fact.

7.1.7 Multi-Bridge Network

The system architecture can be scaled to monitor multiple bridges simultaneously through a centralized cloud dashboard. Each bridge would have its own ESP32 unit and sensor array, all reporting to a common Blynk or custom cloud platform that provides a unified overview of the structural health of an entire regional bridge network.

Conclusion

The Smart Bridge Monitoring and Accident Prevention System successfully demonstrates how modern embedded systems and Internet of Things technology can be applied to a critical and real-world safety challenge. The integration of a simple yet reliable continuity wire collapse detection mechanism with an SW420 vibration sensor provides a dual-layer approach to structural health monitoring that is both effective and affordable.

The ESP32 microcontroller proved to be an excellent choice for this application, offering the computational power needed to run the monitoring firmware, the PWM output capability required for servo motor control, and built-in Wi-Fi for seamless cloud connectivity — all in a single, low-cost package. The Blynk IoT platform provided a user-friendly and responsive interface for remote monitoring and real-time notifications without requiring the development of a custom mobile application.

Testing confirmed that the system responds to dangerous conditions within fractions of a second, closes the safety barrier gate reliably, activates local alarms, and delivers mobile notifications in near real time. The system maintained stable operation over extended test periods without false alarms, demonstrating the practical viability of the design.

From an educational perspective, this project illustrates the practical application of concepts from embedded systems design, sensor interfacing, PWM control, Wi-Fi networking, and IoT platform integration. It serves as a valuable starting point for students and researchers interested in the broader field of smart infrastructure and structural health monitoring.

In conclusion, this project makes a meaningful contribution towards improving bridge safety and reducing the risk of accidents caused by structural failures. With further development along the lines outlined in the Future Scope chapter — including load monitoring, crack detection, solar power, and AI-based analytics — this system has the potential to evolve into a comprehensive solution suitable for deployment on real bridges, ultimately helping to save lives and protect infrastructure.

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